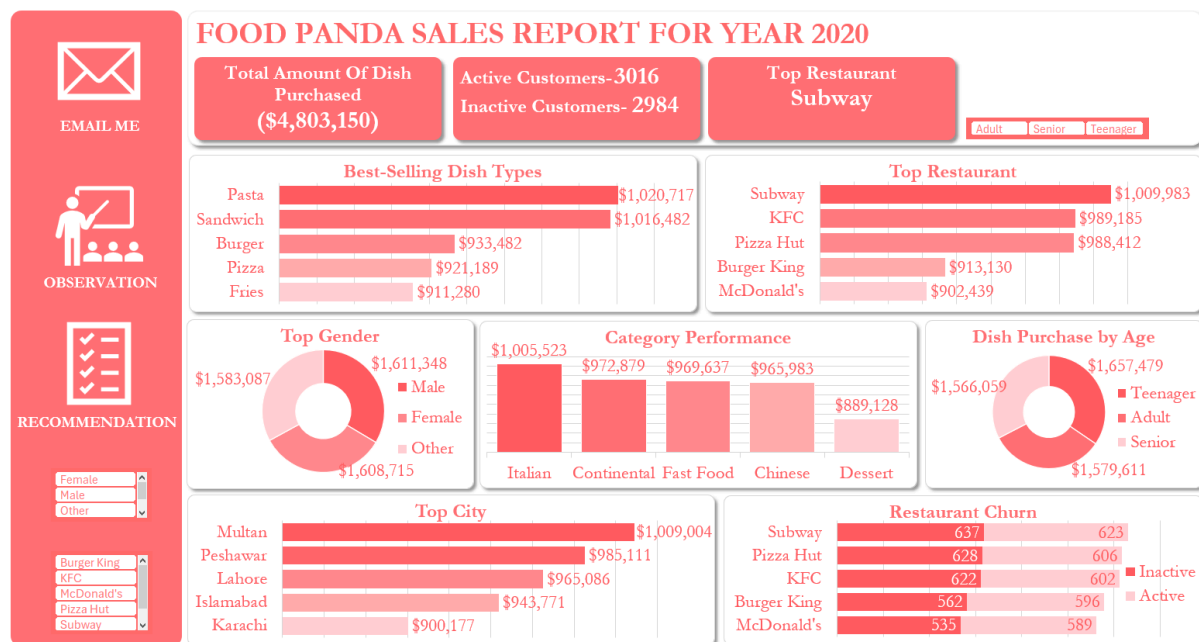


## Food Panda: Restaurant Comparison Analysis



*Food Panda Sales Report for Year 2020.*

### Introduction:

In the culinary industry of today's world, restaurants play a vital role in our dining experience. It has become important for restaurants to understand their strengths, weaknesses and customers' needs to stay ahead in the ever increasing competition in the food industry. This project compares and analyzes the performance of different restaurants in order to identify their strengths and weaknesses. By examining key factors like Restaurant Churn, best-selling dish type, category performance and top restaurant, this study aims to provide insights that can help restaurants improve their business and customer satisfaction.

From the findings of this analysis, restaurant owners, managers and stakeholders can help make informed decisions to drive business growth and success.

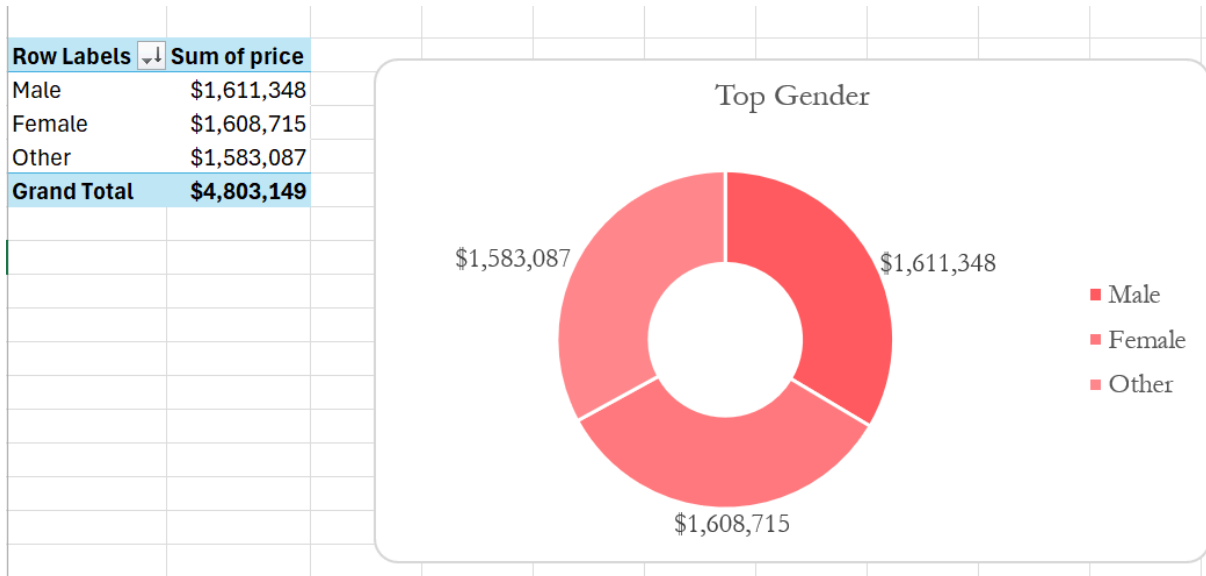
### Objectives:

1. To evaluate and compare the performance of different restaurants.
2. To recognize the strengths and weaknesses of each restaurant.
3. To understand customer preferences and expectations.
4. To develop insights for improving restaurant operations and customer satisfaction.
5. To inform business decisions with data-driven insights.

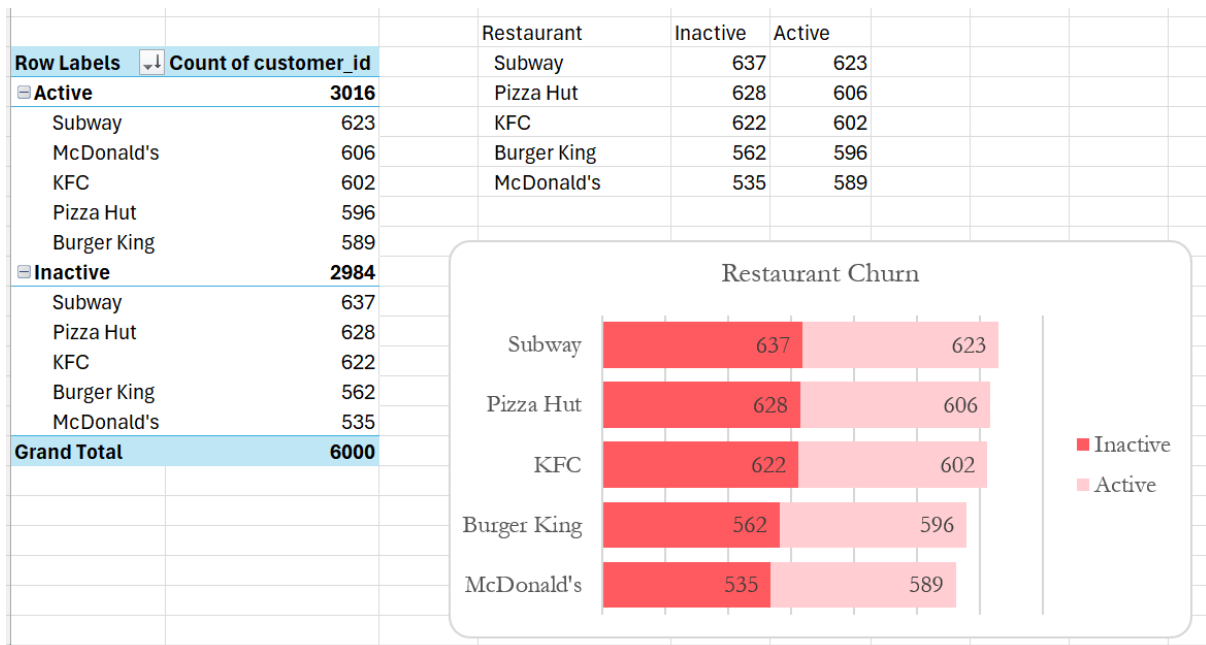
### Methodology:

This project's analysis was carried out using Microsoft Excel. The dataset was cleaned (duplicates were removed, empty cells were checked for) to ensure it is free from errors that could affect the analysis. There was a little shortcoming in the process of cleaning, where an error message was persistently popping up while trying to put the order date column in a text format.

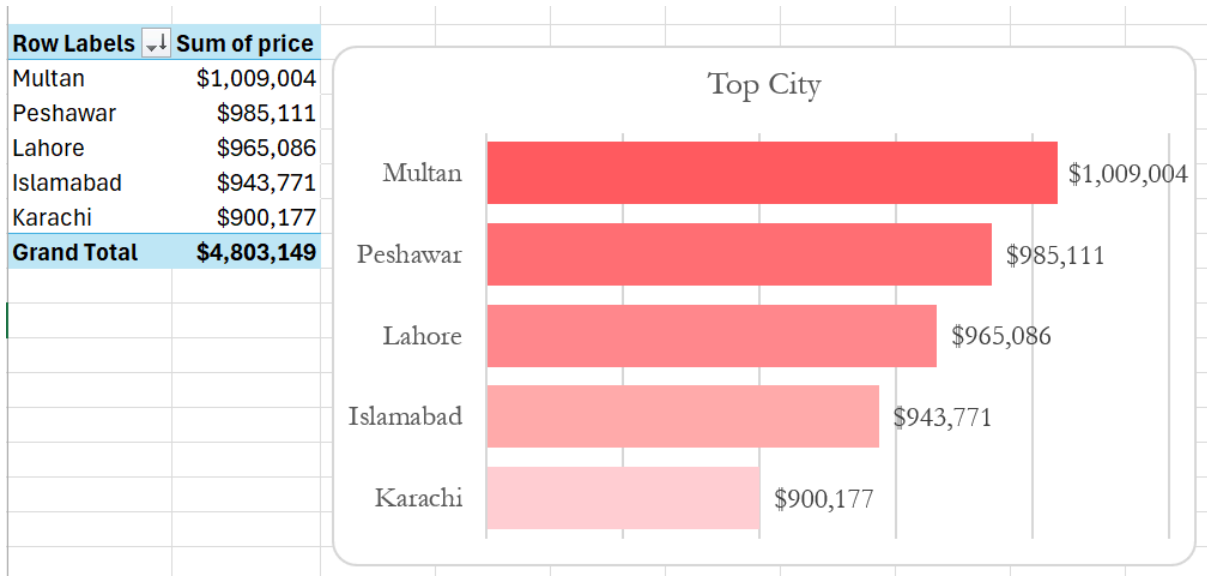
The next step taken for this analysis was to input the data in a pivot table to find the relationship between variables.



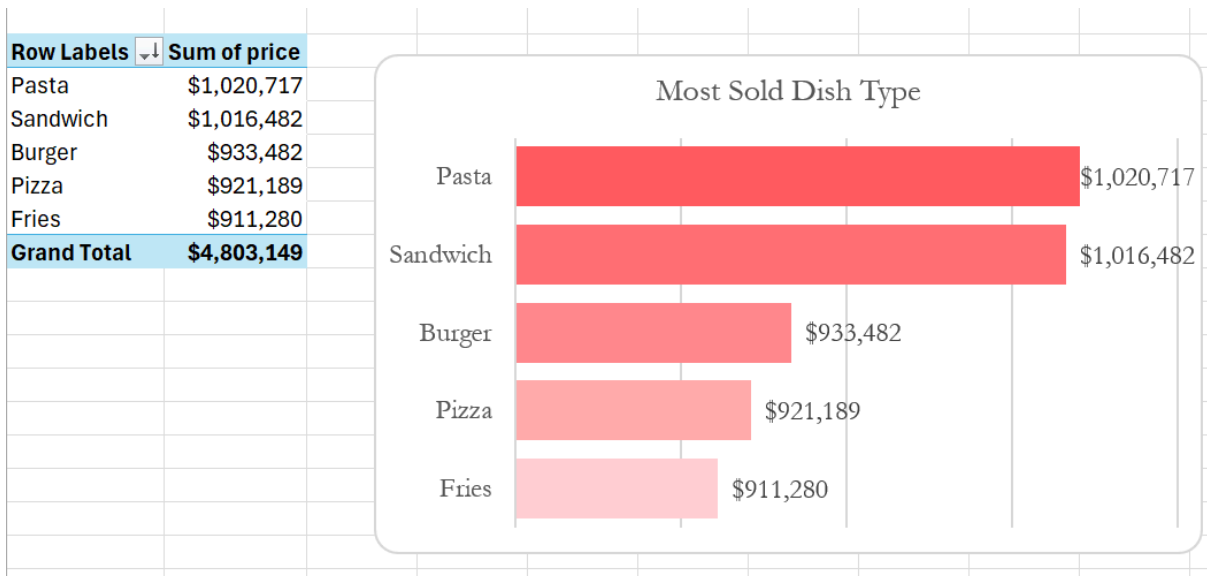
The above diagram is a Pie-chart showing the gender that purchased the most dishes. Here, we have the Male, Female and Other. The male gender topped the chart by purchasing dishes worth \$1,611,348, the Female gender bought dishes worth \$1,608,715 while the gender tagged Other purchased dishes worth \$1,583,087.



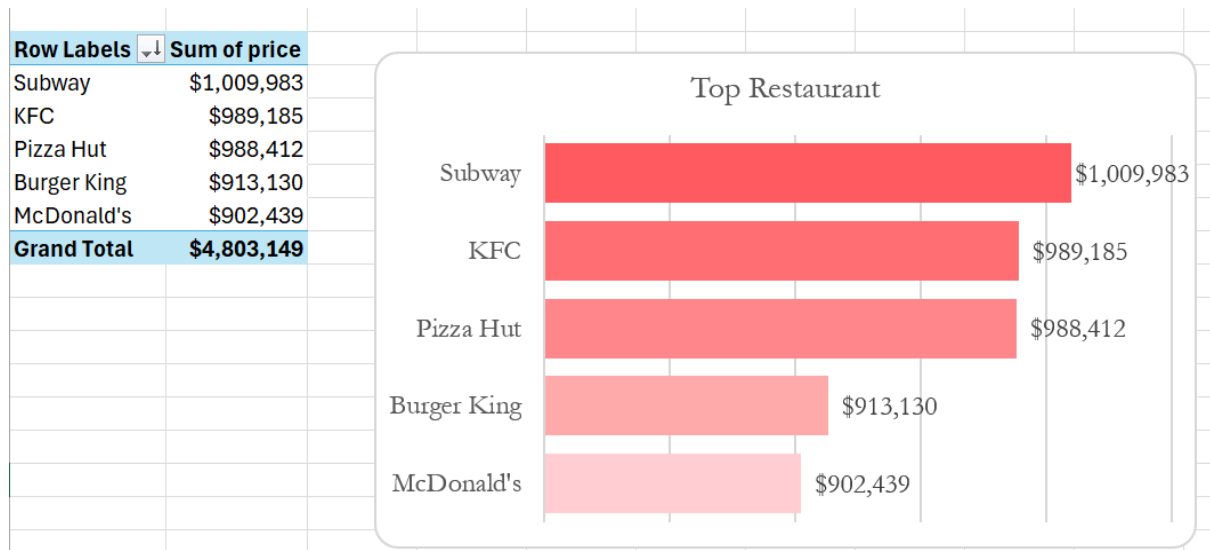
The chart here is a Bar-chart that shows the number of active and inactive customers in 5 different restaurants. Subway restaurant have the highest active customers and also the highest inactive customers while McDonald's has the least active customers and also the least inactive customers. In total, all restaurants have a total of 3016 active customers and 2984 inactive customers.



For the top city, a Bar chart was used as shown above. Each of the restaurants have 5 major cities where their customers come from. From these 5 cities, customers from the City of Multan purchased about \$1,009,004 dishes which is the highest amount, while customers from the City of Karachi purchased the least amount of dishes (\$900,177).



The most sold dish type as shown in the Bar chart above is Pasta. In the 5 different restaurants, Pasta was the most sold dish type. Customers purchased about \$1,020,717 Pasta dishes. Fries was the least purchased dish type with about \$911,280 Fries bought.

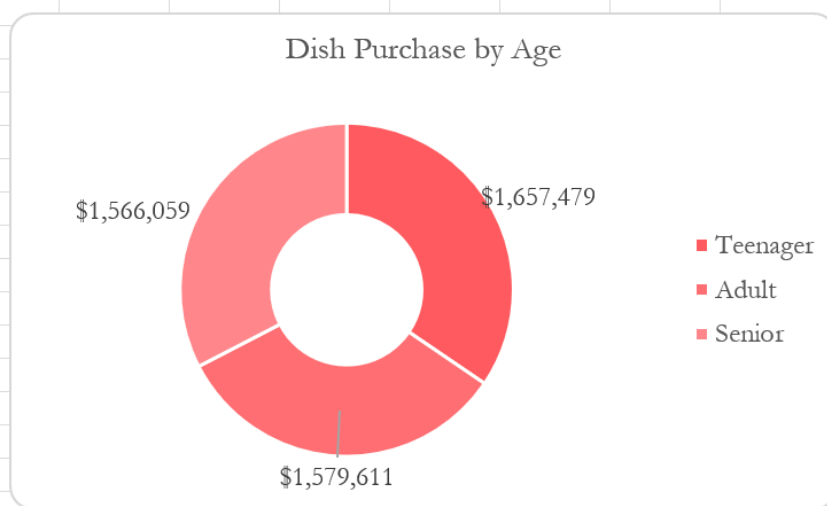


The Top Restaurant as displayed above is Subway Restaurant. This restaurant sold about \$1,009,983 dishes to customers. It also had the highest number of active and inactive customers. Meanwhile, McDonald's sold the least amount of dishes with about \$902,439 dishes sold.



The Food category is where all of the dish types belong. For this category, Italian dishes topped the chart with about \$1,005,523 Italian dishes purchased. Meanwhile, Dessert ranked bottom.

| Row Labels         | Sum of price       |
|--------------------|--------------------|
| Teenager           | \$1,657,479        |
| Adult              | \$1,579,611        |
| Senior             | \$1,566,059        |
| <b>Grand Total</b> | <b>\$4,803,149</b> |



The diagram above is a Pie-chart that shows us the dishes purchased by different Age-groups (Teenagers, Adults and Senior). From the diagram, teenagers purchased the most dishes with about \$1,657,479, Adults purchased dishes worth \$1,579,611 while Seniors purchased dishes worth \$1,566,059.

#### **Story of Data:**

This data tells us about 5 different restaurants, their customer base, customer churn, amount of dishes sold, most patronized restaurants etc. The data paints a picture of different dishes with each restaurant having its own strengths and weaknesses.

The data is a public dataset that was gotten from [kaggle.com](https://www.kaggle.com)

#### **Data Splitting and Preprocessing:**

##### **- Data Cleaning**

The dataset was downloaded from [kaggle.com](https://www.kaggle.com) and loaded into an Excel Workbook. Checks were performed for blanks and duplicates. In order to preserve the original dataset, the raw data was duplicated and retained for reference.

##### **- Data Splitting**

##### **Independent**

- Customer id
- Gender
- Age
- City
- Category
- Restaurant Name
- Dish Name
- Payment method
- Churn
- Order Date
- Order Id

##### **Dependent**

- Price
- Quantity

### **Industry context and Stakeholders**

- The Stakeholders of this project are Restaurant Owners, Manager and Sales Persons

### **Potential-Analysis:**

- Top gender by total price purchased
- Top Category by total price purchased
- Top Restaurant by total price purchased
- Most Sold Dish by total price purchased
- Churn by Restaurant
- Top-leading City by total price purchased
- Dish purchase by Age

### **Potential-Analysis Insights:**

- To find out the gender that purchased the most dish, recognize its most bought dish and make limited time discounts available so as to increase patronage.
- To recognize the top category of dish and ensure it is readily available for customers.
- To finetune out the best restaurant, recognize its best selling measures and apply them to other restaurants.
- To identify the most sold dish type and come up with ways to ensure it maintains its position.
- To identify the number of active and inactive customers in different restaurants and find out the reasons that led to their churn.
- To finetune out the age-group that purchased the most dish and ensure its most purchased dish is readily available for sale.

### **Data Visualizations:**

- Best-Selling Dish: A Bar Chart was used.
- Top Restaurant: A Bar Chart was used.
- Top Gender: A Pie-chart was used.
- Dish Purchase by Age: A Pie-chart was used.
- Category Performance: A Column chart was used.
- Top City: A Bar Chart was used.
- Restaurant Churn: A Stacked-Bar Chart was used.

### **In-Analysis Observations:**

#### **Analysis: Top Gender**

- The Male gender purchased more dishes than the others, with a total purchase price of \$1,611,348.
- The Female gender purchased a total of \$1,608,715 dishes while the gender tagged Other purchased the least (\$1,583,087).

#### **Analysis: Best Category**

- Amongst the 5 categories of food, Italian ranked 1st with a total purchase of \$1,005,523. Continental ranked 2nd with a total of \$972,879, Fast food ranked 3rd with a total of \$969,637.

- Chinese category ranked 4th with a total of \$965,983, while the least is Dessert, with a total of \$889,128.
- There is no huge difference in purchase price between Continental, Fast food and Chinese Category.

#### ***Analysis: Restaurant Churn***

- The total number of Active Customers are 3016, while the total number of inactive customers are 2984.
- Subway Restaurant has the highest number of Active customers (623), next is McDonald's (606), followed by KFC (602), Pizza Hut (596) and Burger King (589) which has the least Active Customers.
- For Inactive Customers, Subway has the highest inactive customers (637), next is Pizza Hut (628), KFC (622), Burger King (562) and McDonald's (535) which has the least inactive customers.
- Subway had the highest Active Customers, but also had the highest Inactive customers.

#### ***Analysis: Top Restaurant***

- Subway ranked top amongst the 5 restaurants. It sold dishes worth \$1,009,983 to its customers. KFC ranked 2nd with a total of \$989,185 dishes sold.
- Pizza Hut ranked 3rd with a total of \$989,412 dishes sold. Burger King ranked 3rd (\$913,130) while McDonald sold the least (\$902,439).
- KFC and Pizza Hut had a slight difference in the total amount of dishes sold.

#### ***Analysis: Top City***

- Customers from Multan purchased the most dishes with a total of \$1,009,004. Peshawar is the 2nd highest with a total of \$985,111. Lahore ranked 3rd with a total of \$965,086 dish purchased, Islamabad ranked 4th with a total of \$943,771 dish purchased while Karachi purchased the least amount of dish (\$900,177).

#### ***Analysis: Top Dish***

- The most sold dish type is Pasta, with a total of \$1,020,717 Pasta purchased. Sandwich ranked 2nd with a total of \$1,016,482 Sandwich purchased, Burger ranked 3rd (\$933,482), Pizza ranked 4th (\$921,189) while Fries was sold the least with a total of \$911,280 Pizza sold.

#### ***Analysis: Dish Purchase by Age***

- Amongst the 3 age-groups, Teenagers bought the most dishes with a total of \$1,657,479. Adults ranked 2nd with a total of \$1,579,611 while Senior purchased the least (\$1,566,059).
- Senior and Adult age-groups spent almost the same amount purchasing different dish-type.

#### ***In-Analysis Recommendations:***

##### ***Analysis: Top Gender***

- Marketing campaigns that will attract females and the other gender should be developed.
- Feedbacks should be collected from females and the other gender so as to understand their tastes and needs.
- Offer discounts that would reach a broader range of customers.

***Analysis: Best Category***

- Mixed and pairing suggestions should be offered to customers to ensure they touch at least 2-3 different categories.
- Flavours and ingredients of every category should be improved upon based on customer feedback and recommendations.

***Analysis: Restaurant Churn***

- Menus should be adjusted to suit customer preferences while loyalty programs are also offered.
- Marketing campaigns should be run so as to reconnect with inactive customers.
- Staff members should be trained on how to attend to the customer base and their complaints.

***Analysis: Top Restaurant***

- Analyze Subway restaurant success factors and apply similar strategies to other restaurants.
- Competitor's pricing should be analyzed and prices should be adjusted while still maintaining profitability.

***Analysis: Top City***

- Feedbacks should be collected from customers in cities with the least patronage so as to understand their needs.
- Limited time discounts should be offered so as to increase patronage.

***Analysis: Top Dish***

- Feedbacks should be collected from customers in cities with the least patronage so as to understand their needs.
- Limited time discounts should be offered so as to increase patronage.

***Analysis: Dish Purchase by Age***

- Family friendly options that appeal to Adults and Seniors should be offered so as to increase patronage.
- Feedbacks should be collected from Seniors and Adults to understand their needs and preferences.
- Discounts should be considered for Seniors and Adults.

***Post-Analysis Observations:******Analysis: Top Gender***

- The Male gender had the highest purchase amongst other genders with a total of \$1,611,348. Male gender from the city of Multan purchased the most dishes than other Male gender in other cities, with a total of \$347,541. This gender purchased Pasta the most (\$344,882), and mostly patronized Pizza Hut Restaurant with dishes worth \$339,631 purchased. By Category, this gender bought the Italian category more (\$347,357) and teenagers purchased more dishes (\$567,312) than other Age group
- The Female gender from the City of Lahore bought more dishes than females from other City with a total of \$344,326. This gender purchased Sandwich the most (\$358,879), and mostly patronized Pizza Hut Restaurant with dishes worth \$350,927 purchased. By



category, this gender bought the Italian category more (\$338,975) and teenagers purchased more dishes (\$550,733) than Seniors and Adults.

- The Other ranked 3rd amongst the 3 genders. This Gender from the City of Multan bought more dishes than those from other City with a total of \$341,799. By category, this gender bought the Fast Food category more (\$330,475) and teenagers purchased more dishes (\$539,434) than Seniors and Adults. This gender purchased Pasta the most (\$326,649), and mostly patronized Subway Restaurant with dishes worth \$358,500 purchased.

### ***Analysis: Best Category***

- The Italian Category topped the chart with a total of \$1,005,523. By gender, the Male gender purchased this category the most with a total of \$347,657. Customers from Lahore purchased this category more than customers from other cities, with a total of \$223,610 Italian dish bought. The most sold dish under this category is Sandwich (\$223,609), while the restaurant that sold this dish the most is Subway restaurant (\$228,902). By age-group, teenagers purchased different dish types from this category than other categories with a total of \$356,466 dish purchased.

- The 2nd best category is Continental, with a total of \$972,879. By gender, the Male gender purchased this category the most with a total of \$328,127. Customers from Multan purchased this category more than customers from other cities, with a total of \$218,018 Continental dish bought. The most sold dish under this category is Sandwich (\$208,710), while the restaurant that sold this dish the most is Subway restaurant (\$217,783). By age-group, Adults purchased different dish types from this category than other categories with a total of \$354,700 dishes purchased.

- The 3rd category by rank is Fast Food, with a total of \$969,637. By gender, the Other gender purchased this category the most with a total of \$330,475. Customers from Multan purchased this category more than customers from other cities, with a total of \$218,134 Fast Food dishes bought. The most sold dish under this category is Pasta (\$210,140), while the restaurant that sold this dish the most is Burger King restaurant (\$208,944). By age-group, seniors purchased different dish types from this category than other categories with a total of \$342,333 dishes purchased.

- The 4th category by rank is Chinese, with a total of \$965,983. By gender, the Male gender purchased this category the most with a total of \$341,168. Customers from Lahore purchased this category more than customers from other cities, with a total of \$214,064 Chinese dishes bought. The most sold dish under this category is Pasta (\$212,704), while the restaurant that sold this dish the most is Pizza Hut restaurant (\$220,974). By age-group, adults purchased different dish types from this category than other categories with a total of \$341,100 dishes purchased.

- The lowest category by rank is Dessert, with a total of \$889,128. By gender, the Female gender purchased this category the most with a total of \$309,234. Customers from Peshawar purchased this category more than customers from other cities, with a total of \$197,375 Dessert dish bought. The most sold dish under this category is Sandwich (\$190,316), while the restaurant that sold this dish the most is Pizza Hut restaurant (\$193,696). By age-group, Teenagers purchased different dish types from this category than other categories with a total of \$336,385 dish purchased.

### ***Analysis: Restaurant Churn***

- The total number of customers is 6000. 3016 are Active while 2984 are inactive.

- Subway Restaurant has the highest number of Active Customers (623) and also the highest number of Inactive Customers (637). By gender, it has 212 Active and 206 Inactive Female, 199 Active and 216 Inactive Male, 212 Active and 215 Inactive Other gender. By Age, it has 209 Active and 209 Inactive Adults, 189 Active and 210 Inactive Senior, 225 Active and 218 Inactive Teenagers. By city, it has 127 Active & 115 Inactive Customers in Islamabad, 117 Active & 113 Inactive Customers in Karachi, 126 Active & 138 Inactive Customers in Lahore, 131 Active & 154 Inactive Customers in Multan, 122 Active & 117 Inactive Customers in Peshawar.

- Pizza Hut Restaurant has 596 Active Customers and 628 Inactive Customers. By gender, it has 217 Active and 217 Inactive Female, 202 Active and 218 Inactive Male, 177 Active and 193 Inactive Other gender. By Age, it has 200 Active and 202 Inactive Adults, 200 Active and 210 Inactive Senior, 196 Active and 216 Inactive Teenagers. By city, it has 103 Active & 113 Inactive Customers in Islamabad, 121 Active & 114 Inactive Customers in Karachi, 123 Active & 134 Inactive Customers in Lahore, 118 Active & 141 Inactive Customers in Multan, 131 Active & 126 Inactive Customers in Peshawar.

- KFC Restaurant has 602 Active Customers and 622 Inactive Customers. By gender, it has 196 Active and 217 Inactive Female, 202 Active and 200 Inactive Male, 199 Active and 226 Inactive Other gender. By Age, it has 174 Active and 202 Inactive Adults, 193 Active and 207 Inactive Seniors, 235 Active and 213 Inactive Teenagers. By city, it has 132 Active & 139 Inactive Customers in Islamabad, 123 Active & 115 Inactive Customers in Karachi, 102 Active & 124 Inactive Customers in Lahore, 124 Active & 131 Inactive Customers in Multan, 121 Active & 113 Inactive Customers in Peshawar.

- Burger King Restaurant has 589 Active Customers and 562 Inactive Customers. By gender, it has 197 Active and 178 Inactive Female, 203 Active and 190 Inactive Male, 189 Active and 194 Inactive Other gender. By Age, it has 214 Active and 182 Inactive Adults, 184 Active and 195 Inactive Seniors, 191 Active and 185 Inactive Teenagers. By city, it has 126 Active & 112 Inactive Customers in Islamabad, 116 Active & 114 Inactive Customers in Karachi, 111 Active & 115 Inactive Customers in Lahore, 110 Active & 111 Inactive Customers in Multan, 126 Active & 110 Inactive Customers in Peshawar.

- McDonald's Restaurant has 606 Active Customers and 535 Inactive Customers. By gender, it has 200 Active and 194 Inactive Female, 203 Active and 184 Inactive Male, 203 Active and 157 Inactive Other gender. By Age, it has 224 Active and 168 Inactive Adults, 183 Active and 183 Inactive Senior, 199 Active and 184 Inactive Teenagers. By city, it has 114 Active & 106 Inactive Customers in Islamabad, 120 Active & 92 Inactive Customers in Karachi, 124 Active & 120 Inactive Customers in Lahore, 125 Active & 111 Inactive Customers in Multan, 123 Active & 106 Inactive Customers in Peshawar.

### ***Analysis: Top Restaurant***

- Subway topped the Chart amongst other restaurants with a total of \$1,009,983. Its most sold dish type is Pasta, with a total of \$226,304 Pasta sold. Amongst the 3 gender, the gender tagged Other is the highest patronizer of Subway restaurant (\$358,500). Subway has 623 Active customers and 637 Inactive customers. The Customers from Multan patronized this restaurant the most, they purchased dishes worth \$233,413. By Category, the Italian Category was sold the most with a total of \$228,902, while by Age-group, the Teenagers patronized this restaurant the most than other age-groups with dishes worth \$354,565.

- The 2nd best restaurant is KFC with a total of \$989,185. Its most sold dish type is Sandwich, with a total of \$218,272 Sandwiches sold. Amongst the 3 gender, the gender

tagged Other is the highest patronizer of KFC restaurant (\$341,014). KFC has 602 Active customers and 622 Inactive customers. The Customers from Islamabad patronized this restaurant the most, they purchased dishes worth \$220,590. By Category, the Chinese Category was sold the most with a total of \$207,959, while by Age-group, the Teenagers patronized this restaurant the most than other age-groups with dishes worth \$372,478.

- The 3rd restaurant on the Chart is Pizza Hut with a total of \$988,412. Its most sold dish type is Pasta, with a total of \$224,529 Pasta sold. Amongst the 3 gender, Female gender is the highest patronizer of Pizza Hut restaurant (\$350,927). Pizza Hut has 596 Active customers and 628 Inactive customers. The Customers from Pulshawar patronized this restaurant the most, they purchased dishes worth \$212,282. By Category, the Chinese Category was sold the most with a total of \$220,974, while by Age-group, the Teenagers patronized this restaurant the most than other age-groups with dishes worth \$334,822.

- The 4th restaurant on the chart is Burger King with a total of \$913,130. Its most sold dish type is Sandwich, with a total of \$203,951 Sandwiches sold. Amongst the 3 gender, the Female Gender is the highest patronizer of Burger King restaurant (\$305,549). Burger King has 589 Active customers and 562 Inactive customers. The Customers from Pulshawar patronized this restaurant the most, they purchased dishes worth \$198,401. By Category, the Fast Food Category was sold the most with a total of \$208,944, while by Age-group, the Senior patronized this restaurant the most than other age-groups with dishes worth \$307,644.

- The least restaurant by rank is McDonald's with a total of \$902,439. Its most sold dish type is Burger, with a total of \$198,470 Burger sold. Amongst the 3 gender, the Male Gender is the highest patronizer of McDonald's restaurant (\$311,809). McDonald's has 606 Active customers and 535 Inactive customers. The Customers from Lahore patronized this restaurant the most, they purchased dishes worth \$187,267. By Category, the Italian Category was sold the most with a total of \$191,769, while by Age-group, the Adults patronized this restaurant the most than other age-groups with dishes worth \$319,665.

### ***Analysis: Top City***

- The city with the most customers is Multan at \$1,009,004. The Male gender from this City purchased the most items with a total of \$347,541. By dish type, customers from this City purchased Burger the most (\$210,029) while they patronized Subway restaurant (\$233,413) more than other restaurants. The most purchased category by customers from this City is the Italian Category, with a total of \$291,091 while Teenagers in this City purchased more dishes (\$357,019) than other age-groups.

- Next is Peshawar \$985,111. The Male gender from this City purchased the most items with a total of \$333,053. By dish type, customers from this City purchased Pasta the most (\$222,682) while they patronized Pizza Hut restaurant (\$212,282) more than other restaurants. The most purchased category by customers from this City is the Chinese Category, with a total of \$203,640 while Senior in this City purchased more dishes (\$341,949) than other age-groups.

- Next is Lahore \$965,086. The Male gender from this City purchased the most items with a total of \$344,326. By dish type, customers from this City purchased Pasta the most (\$212,480) while they patronized Subway restaurant (\$207,051) more than other restaurants. The most purchased category by customers from this City is the Italian Category, with a total of \$223,610 while Adults in this City purchased more dishes (\$333,757) than other age-groups.

- The 4th City by rank is Islamabad \$943,771. The Male gender from this City purchased the most items with a total of \$321,200. By dish type, customers from this City purchased Fries the most (\$193,844) while they patronized KFC restaurant (\$220,590) more than other restaurants. The most purchased category by customers from this City is the Italian Category, with a total of \$196,614 while Teenagers in this City purchased more dishes (\$333,778) than other age-groups.
- The least City by rank is Karachi at \$900,177. The Other gender from this City purchased the most items with a total of \$305,315. By dish type, customers from this City purchased Sandwich the most (\$211,383) while they patronized Pizza Hut restaurant (\$185,400) more than other restaurants. The most purchased category by customers from this City is the Fast Food Category, with a total of \$200,825 while Adults in this City purchased more dishes (\$306,524) more than other age-groups.

### ***Analysis: Top Dish***

- The best-selling dish type is Pasta, with a total of \$1,020,717 Pasta Sold. Pasta under the Italian Category was sold the most than the Pasta in other Categories (\$214,497). Subway restaurant sold Pasta the most amongst other restaurants with a total of \$226,304. The Customers from the City of Peshawar purchased Pasta the most with a total of \$222,682. Amongst the 3 Gender, the Female Gender purchased Pasta the most with a total of \$349,187 while amongst Age-Group, teenagers purchased Pasta the most with a total of \$353,127.
- The 2nd best-selling dish type is Sandwich, with a total of \$1,016,482 Sandwiches Sold. Sandwich under the Italian Category was sold the most than the Sandwich in other Categories (\$223,609). KFC restaurant sold sandwiches the most amongst other restaurants with a total of \$218,272. The Customers from the City of Peshawar purchased Sandwich the most with a total of \$213,017. Amongst the 3 Gender, the Female Gender purchased Sandwich the most with a total of \$358,879 while amongst Age-Group, Senior purchased Sandwich the most with a total of \$348,648.
- The 3rd best-selling dish type is Burger, with a total of \$933,482 Burger Sold. The burger under the Italian Category was sold the most than the Burger in other Categories (\$199,851). McDonald's restaurant sold burgers the most amongst other restaurants with a total of \$198,470. The Customers from the City of Multan purchased Burger the most with a total of \$210,029. Amongst the 3 Gender, the Male Gender purchased Burger the most with a total of \$319,534 while amongst Age-Group, Adult purchased Burger the most with a total of \$312,759.
- The 4th best-selling dish type is Pizza, with a total of \$921,189 Pizzas Sold. Pizza under the Fast Food Category was sold the most than the Pizza in other Categories (\$212,323). Subway restaurant sold Pizza the most amongst other restaurants with a total of \$212,823. The Customers from the City of Peshawar purchased Pizza the most with a total of \$198,263. Amongst the 3 Gender, the Male Gender purchased Pizza the most with a total of \$311,860 while amongst Age-Group, Teenagers purchased Pizza the most with a total of \$335,552.
- The least sold dish type is Fries, with a total of \$911,280 Fries Sold. Fries under the Chinese Category were sold the most than the Fries in other Categories (\$192,179). Subway restaurant sold Fries the most amongst other restaurants with a total of \$200,611. The Customers from the City of Multan purchased Fries the most with a total of \$206,379. Amongst the 3 Gender, the Other Gender purchased Fries the most with a total of

\$319,238 while amongst Age-Group, Teenagers purchased Fries the most with a total of \$321,301.

### ***Analysis: Dish Purchase by Age***

- By Age, teenagers purchased more dishes than Seniors and Adults with a total of \$1,657,479. By Category, Teenagers purchased Italian category the most with a total of \$356,466 dish purchased. They patronized KFC the most, buying dishes worth \$372,478 and for the dish type, they purchased sandwiches the most (\$353,127). Teenagers from Multan bought more dishes than teenagers from other cities, with a total of \$357,019 dish purchased. 218 Teenagers churned Subway restaurant, 184 churned McDonald's, 213 churned KFC, 216 churned Pizza Hut while 185 churned Burger King. Moreover, 225 teenagers are active in Subway Restaurant, 196 in Pizza Hut, 235 in KFC, 191 in Burger King and 199 in McDonald's. This shows that Subway has the highest number of Churned/inactive teenagers while KFC has the highest number of Active Teenagers.
- Adult ranked 2nd with a total of \$1,579,611. By category, adults purchased Continental category the most with a total of \$354,700 dishes purchased. They patronized Subway the most, buying dishes worth \$342,773 and for dish type, they purchased Sandwich the most (\$330,794). Adults from Lahore bought more dishes than Adults from other cities, with a total of \$333,757 dish purchased. 209 Adults churned Subway restaurant, 168 churned McDonald's, 202 churned KFC, 202 churned Pizza Hut while 182 churned Burger King. Moreover, 209 Adults are active in Subway Restaurant, 200 in Pizza Hut, 174 in KFC, 214 in Burger King and 224 in McDonald's. This shows that Subway has the highest number of Churned/inactive Adults while McDonald's has the highest number of Active Adults.
- Senior ranked the least amongst the 3 Age-groups with a total of \$1,566,059. By Category, Senior purchased the Fast Food category the most with a total of \$342,333 dish purchased. They patronized Pizza Hut the most, buying dishes worth \$331,108 and for dish type, they purchased sandwiches the most (\$348,648). Seniors from Multan bought more dishes than Adults from other cities, with a total of \$349,940 dishes purchased. 210 Senior churned Subway restaurant, 183 churned McDonald's, 207 churned KFC, 210 churned Pizza Hut while 195 churned Burger King. Moreover, 189 Seniors are active in Subway Restaurant, 200 in Pizza Hut, 193 in KFC, 184 in Burger King and 183 in McDonald's. This shows that Subway and Pizza Hut have the highest number of Churned/inactive Senior while Pizza Hut has the highest number of Active Senior.

### ***Post-Analysis Insights:***

#### ***Analysis: Top Gender***

- Marketing campaigns that will attract females and the other gender should be developed.
- Feedbacks should be collected from females and the other gender so as to understand their tastes and needs.
- Offer discounts that would reach a broader range of customers.

#### ***Analysis: Best Category***

- Staff members should be educated on different categories so they can improve on their recommendation skills to customers.
- The Italian category's success factor should be analyzed and applied to other categories so as to replicate its success.
- Mixed and pairing suggestions should be offered to customers to ensure they touch at least 2-3 different categories.

- Flavours and ingredients of every category should be improved upon based on customer feedback and recommendations.

#### ***Analysis: Restaurant Churn***

- The overall total number of inactive customers is more than the number of Active customers, this shows that almost 60% of the customer base is no longer engaging with the restaurants. This may be due to customer experience, new competitors, or change in preference of customers.
- Feedbacks should be collected from customers to know areas of improvement and make necessary adjustments.
- Menus should be adjusted to suit customer preferences while loyalty programs are also offered.
- Marketing campaigns should be run so as to reconnect with inactive customers.
- Staff members should be trained on how to attend to the customer base and their complaints.

#### ***Analysis: Top Restaurant***

- Social media should be leveraged to advertise each of the restaurants so as to attract new customers.
- Analyze Subway restaurant success factors and apply similar strategies to other restaurants.
- Competitor's pricing should be analyzed and prices should be adjusted while still maintaining profitability.
- Quality food, consistent taste and good presentation of the dish should be maintained.
- Influencers can be partnered with so as to increase visibility of each restaurant.

#### ***Analysis: Top City***

- To build a customer base in other cities, the menu should be adjusted to suit local taste and preferences.
- There should be efficient delivery services to reach customers in all cities.
- Feedbacks should be collected from customers in cities with the least patronage so as to understand their needs.
- Limited time discounts should be offered so as to increase patronage.

#### ***Analysis: Top Dish***

- Pairing suggestions should be offered to customers. It could be pairing other dishes to popular dishes like Pasta so as to ensure every other dish is purchased.
- Pricing and portion sizes should be adjusted while still maintaining profitability.
- Customer feedback regarding other dishes should be collected.
- Every dish should be made appealing while still maintaining a good customer relationship.
- Staff members should be trained on dish preparation, pairing suggestions and ingredients.

#### ***Analysis: Dish Purchase by Age***

- Family friendly options that appeal to Adults and Seniors should be offered so as to increase patronage.

- Feedbacks should be collected from Seniors and Adults to understand their needs and preferences.
- Discounts should be considered for Seniors and Adults.
- Dishes that cater to different age groups should be considered in the menu.

**Conclusion:**

This analysis of 5 different restaurants showcases the importance of understanding customer interests and tailoring offerings to meet their needs. By leveraging these insights, restaurants can improve customer satisfaction, drive loyalty, and ultimately, boost their business.

**Reference:** [kaggle.com](https://www.kaggle.com)